

Efficient Multi-View 3D Human Pose Estimation via Knowledge Distillation

Wei Herng Yap*, Rui Cao*, Sim Kuan Goh

School of Electrical Engineering and Artificial Intelligence,
Xiamen University Malaysia, Selangor, Malaysia.

*Contributed equally to this work.

Abstract—In this paper, we focus on the challenge that arises when deploying 3D HPE models in real-world scenarios where computational resources and memory constraints are prevalent. To address this challenge, our research focuses on leveraging knowledge distillation as a technique to reduce the model's complexity and computational demands while preserving its performance. Knowledge distillation involves transferring the knowledge acquired by a complex teacher model to a more lightweight student model. By distilling the essential information, we aim to create a streamlined and efficient 3D HPE model capable of maintaining high accuracy with reduced computational overhead. By training the Student Model with different types of ResNet model, i.e., ResNet-50, ResNet-101, we get the result that with the rate of declining parameters from Teacher Model to Student Model reaching 41% for ResNet-50 and 73% for ResNet-101, it's able to conclude we address the challenge of deploying 3D HPE systems in resource-constrained real-world settings. Ultimately, we aspire to empower researchers and practitioners with adaptable, efficient tools that transcend the boundaries of 3D pose estimation, shaping the landscape of computer vision applications.

Index Terms—Multi-view 3D Human Pose Estimation, Knowledge Distillation

I. INTRODUCTION

In recent years, 3D Human Pose Estimation (3D HPE) has emerged as a pivotal task in computer vision with applications ranging from human-computer interaction to augmented reality [1]–[3], [10]. The accurate estimation of human poses in three-dimensional space plays a crucial role in understanding and interpreting human activities. However, achieving state-of-the-art performance in 3D HPE often demands deep and complex neural network architectures, resulting in models with a high number of parameters, hindering real-time applications.

The challenge arises when deploying such models in real-world scenarios where computational resources and memory constraints are prevalent [15]. To address this challenge, our research focuses on leveraging knowledge distillation as a technique to reduce the model's complexity and computational demands while preserving its performance. Knowledge distillation involves transferring the knowledge acquired by

a complex teacher model to a more lightweight student model [16]. By distilling the essential information, we aim to create a streamlined and efficient 3D HPE model capable of maintaining high accuracy with reduced computational overhead. Also, achieving high accuracy in HPE and 3D HPE is essential for their effectiveness in real-world applications. However, various factors can affect the accuracy of these models, including occlusions, variations in human poses, and different environmental conditions. Occlusions, where parts of the body are not visible in the input images, pose a significant challenge as the model must infer the positions of hidden joints. Additionally, ensuring consistent performance across diverse datasets and real-world scenarios is crucial for generalizing the models' capabilities beyond controlled environments. In addition, in realm of 3DHPE, how to accurately capturing and interpreting depth information is a key challenge. Depth estimation requires precise data from sensors or cameras, and any inaccuracies in this data can lead to significant errors in pose estimation. Depth-related challenges also include integrating data from multiple viewpoints and dealing with the inherent noise in depth sensors [7]. Developing robust methods to handle these issues is essential for improving the reliability of 3D pose estimation.

This paper presents our exploration into the application of knowledge distillation for multi-view 3D human pose estimation, with the primary goal of achieving a model that not only meets the stringent resource constraints of real-world deployment but also serves as a valuable tool in scenarios where computational efficiency is paramount. We delve into the methodology, experimental setup, and results, showcasing the potential of our approach in reducing the training threshold and deploying cost of training the model.

Our research contributions centered on applying knowledge distillation on multi-view 3D human pose estimation make noteworthy contributions to the field and beyond. (1) By streamlining models through knowledge distillation, we address the challenge of deploying 3D HPE systems in resource-constrained real-world settings. (2) The methodology not only lowers training thresholds but also provides a template for resource-efficient solutions in various computer vision domains. (3) Additionally, our work contributes to advancing model generalization, ensuring robust performance across diverse datasets and scenarios. Ultimately, we aspire to empower researchers and practitioners with adaptable, efficient tools that

This research is supported by the Ministry of Higher Education Malaysia through the Fundamental Research Grant Scheme (FRGS/1/2023/ICT02/XMU/02/1), and Xiamen University Malaysia through Xiamen University Malaysia Research Fund (XMUMRF/2022-C10/IECE/0039 and XMUMRF/2024-C13/IECE/0049). Corresponding: simkuan.goh@xmu.edu.my

979-8-3503-7728-6/24/\$31.00 ©2024 IEEE

transcend the boundaries of 3D pose estimation, shaping the landscape of computer vision applications.

To provide an overview of the entire paper, our project consists of the following sections: (1) Related work with three main aspects which are 3DHPE, Model Compression and Knowledge Distillation. (2) Methodology with our algorithms of Knowledge distillation (KD) model followed by (3) Experiment filed illustrating datasets specifications, training, result and analysis of the experiment. Then it comes with (4) conclusion part to have a brief summary. At last, it will be the (5) limitations and potential future development of our work.

II. RELATED WORK

The landscape of Human Pose Estimation (HPE). HPE can be characterized by diverse methodologies, such as regression methods, heatmap-based methods (top-down approach), and bottom-up approach. Regression Methods simplify the problem by mapping human instances directly to skeletal structures, focusing on predicting precise coordinates of keypoints [1], [4]. In contrast, heatmap-Based Methods generate probability heatmaps for keypoints, providing a more representative and widely adopted strategy [5]. The Bottom-Up Approach tackles multiple single-person pose estimations (SPPE), though it may face challenges with occlusions [6], [7]. Finally, the Top-Down Approach locates keypoints first and assigns them to individuals, demonstrating a structured, sequential process [8], [9]. This comprehensive categorization sets the stage for our work, leveraging knowledge distillation to enhance the efficiency and deployability of 3D Human Pose Estimation models.

3D Human Pose Estimation. In 3D HPE, existing works can be categorized into two main approaches, which are the single-stage approach and the two-stage approach. A single-stage methodology involves the direct regression of 3D pose from input images [10], [12], [14]. This approach demands an extensive dataset of image-pose pairs and robust computing resources due to the inherent complexity of directly inferring 3D poses from raw images. In contrast, the two-stage paradigm [11], [13] encompasses the following steps: Utilizing off-the-shelf 2D pose detectors to estimate the coordinates of 2D joints from input images and employing a network model to lift the obtained 2D coordinates into 3D space, facilitating a comprehensive understanding of the three-dimensional pose.

These two distinct approaches offer alternative strategies for tracking the challenging task of 3D pose estimation. While the single-stage approach necessitates a significant amount of labeled data and computational power, the two-stage approach leverages pre-existing 2D pose detectors to enhance the accuracy of 3D pose estimation.

For complexity analysis, the single-stage approach and the two-stage approach both have their own advantages as well as disadvantages. In short, 1) The single-stage approach offers simplicity in modeling but demands a substantial amount of data and computational power. 2) The two-stage approach, while potentially reducing the data requirement, introduces complexities in the form of error propagation and additional

processing steps. 3) The choice between these approaches depends on the available resources, the size and quality of the annotated dataset, and the desired trade-off between model simplicity and computational demand.

Model Compression approaches. Efforts to enhance the efficiency of neural networks have led to the exploration of model compression techniques. Several pivotal papers introduce distinctive approaches, namely pruning, trained quantization, and knowledge distillation.

- **Pruning:** It reduces the number of connections of the network [17], which can learn connection importance, leading to the removal of redundant connections, followed by fine-tuning retained connections. Demonstrating effectiveness on networks like AlexNet and VGG-16, significant parameter reduction (9x and 13x) is achieved without compromising accuracy [16].
- **Quantization:** It enforces weight sharing. Cooperating with Huffman coding, which takes advantage of the biased distribution of effective weights, the comprehensive method achieves substantial storage reduction (35x to 49x) in major neural networks without sacrificing too much accuracy [16], [17].
- **Knowledge Distillation:** Training a large ensemble of models and distilling their knowledge into a single model can improve performance. Distillation can be achieved by raising the temperature of the softmax output layer of the cumbersome model and using the same high temperature when training the small model [15].

Knowledge Distillation. Addressing computational challenges, knowledge distillation transfers the knowledge from a relatively large model to a relatively smaller models while having a lower computational complexity with minor sacrificing on the accuracy. One application [19] of KD is to compress ensemble of machine learning models into a single deployable model, this technique significantly improves performance on tasks such as MNIST and speech recognition. Distillation can be done using a transfer set of unlabeled or labeled data, and adding a small term to the objective function that encourages the small model to predict the true targets. What's more, the effectiveness of distillation depends on the size of the distilled model and the temperature used during training, and matching the logits (inputs to the softmax) of the cumbersome model is a special case of distillation. [18]. Moreover, there are three most common kinds of distillation, which are Online Distillation, Offline Distillation, and Self-Distillation.

- **Online Distillation:**
In online distillation, both the teacher model and the student model are updated simultaneously, and the whole knowledge distillation framework is end-to-end trainable.
- **Offline Distillation:**
In the industrial setting, offline distillation is commonly employed, where students learn from pre-trained teachers. This approach is straightforward and easy to implement. The primary focuses of offline distillation lie in selecting knowledge acquisition methods and designing loss func-

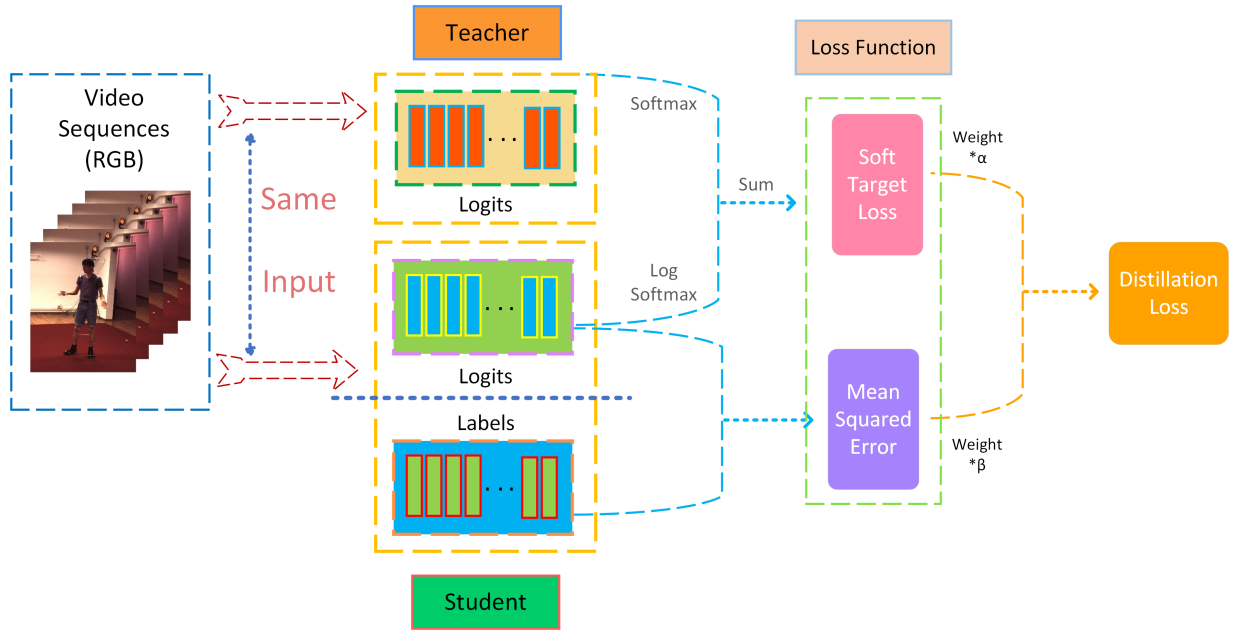


Fig. 1. The Teacher Model and Student Model use the same image input from the Human3.6 dataset, and both generate outputs by the logits unit. The output of Teacher Model - O_T performs the Softmax function, and meanwhile, the output of Student Model - O_S performs the Log-Softmax function, which forms the Soft Target Loss (L_{ST}) - the sum of O'_T and O'_S . Then, we estimate the accuracy and efficiency of the Student Model through Mean Squared Error (L_{MSE}) by O_S and true labels. Finally, we let L_{ST} multiply with weight α and let L_{MSE} multiply with weight β so that the sum of the two forms Distillation Loss, which illustrates the performance of the Knowledge Distillation.

tions. However, a notable challenge in offline distillation arises from the existence of a model capacity gap between the large teacher and the smaller student.

- Self-Distillation:

In Self-Distillation, the same networks are used for the teacher and student models.

What's more, some existing work, such as that by Hinton et al. [15] on KD, has shown that it is possible to transfer the knowledge from a large teacher model to a smaller student model with minimal loss in performance. We extend this concept to the domain of 3D HPE by using multi-view input data and complex 3D pose estimation models as teachers to train more efficient student models. We challenge the traditional approach of using highly complex models for 3D HPE by applying KD to achieve a balance between model size and performance. Unlike pruning and quantization, which primarily focus on reducing model size and inference time, our KD approach ensures that the distilled student model retains critical information and generalizes well across different datasets. In short, Offline distillation means the knowledgeable teacher teaches a student knowledge; online distillation is where both teacher and student study together with each other; self-distillation let student learn knowledge by itself.

III. METHOD

In our Knowledge Distillation (KD) model, we operate within a framework where both the Teacher Model and Student Model utilize the same set of images, denoted as $I = \{I_1, I_2, I_3, \dots, I_n\}$, sourced from the Human3.6 dataset.

The logits unit of each model is employed for predictions. Subsequently, the outputs of both the Teacher Model, denoted as $O_T = \{O_{T_1}, O_{T_2}, O_{T_3}, \dots, O_{T_n}\}$, and the Student Model, denoted as $O_S = \{O_{S_1}, O_{S_2}, O_{S_3}, \dots, O_{S_n}\}$, undergo activation functions for further processing.

The Teacher Model output, O_T , undergoes the Softmax activation function, defined as:

$$\text{Softmax}(O_{T_i}) = \frac{e^{O_{T_i}}}{\sum_{j=1}^n e^{O_{T_j}}}$$

Conversely, the Student Model output, O_S , undergoes the Log-Softmax function, which computes the natural logarithm of the softmax probabilities:

$$\text{Log-Softmax}(O_{S_i}) = \log \left(\frac{e^{O_{T_i}}}{\sum_{j=1}^n e^{O_{T_j}}} \right)$$

We here denote the outputs of Softmax function with teacher's logits and Log-Softmax function with student's logits as O'_T and O'_S respectively, contribute to a method termed Soft Target Loss (L_{ST}), computed as the sum of O'_T and O'_S :

$$L_{ST} = \frac{1}{n} \sum_{i=1}^n (O'_{T_i} + O'_{S_i})$$

The evaluation of Soft Target Loss, ranging from $(-\infty, 1]$, provides insights into the performance of the two models. A value closer to 1 indicates the efficiency and accuracy of Knowledge Distillation, particularly in the context of light-weight models for Human Pose Estimation (HPE).

Furthermore, we perform Mean Squared Error (L_{MSE}) using two components: the original output of the Student Model, O_S , and the true labels of the input images from the dataset, denoted as O_L . The Mean Squared Error Loss is calculated as follows:

$$L_{\text{MSE}} = \frac{1}{n} \sum_{i=1}^n (O_{L_i} - \hat{O}_{S_i})^2$$

After computing the Soft Target Loss and Mean Squared Error, we introduce the concept of Distillation Loss ($L_{\text{Distillation}}$), resulting from a combination of Soft Target Loss and Mean Squared Error Loss:

$$L_{\text{Distillation}} = \alpha \cdot L_{\text{SL}} + \beta \cdot L_{\text{MSE}}$$

This is the loss where the back-propagation will use as the weight updating loss for the models to learn. Here, α and β act as hyperparameters, serving as weighting factors in the loss calculation. α controls the weight of the students models learning from the the teacher models, while β is the weight that controls the student models learning from its own error.

Algorithm 1 Knowledge Distillation Algorithm

```

1: Inputs:
2:   Teacher Model logits:  $O_T$ 
3:   Student Model logits:  $O_S$ 
4:   True labels from dataset:  $O_L$ 
5:   Hyperparameters:  $\alpha, \beta$ 
6:   Number of samples:  $n$ 
7: Initialize Soft Target Loss  $L_{\text{ST}} \leftarrow 0$ 
8: Initialize Mean Squared Error Loss  $L_{\text{MSE}} \leftarrow 0$ 
9: function SOFTMAX(logits)
10:   exp_logits  $\leftarrow \exp(\text{logits})$ 
11:   sum_exp_logits  $\leftarrow \sum \text{exp\_logits}$ 
12:   return  $\frac{\text{exp\_logits}}{\text{sum\_exp\_logits}}$ 
13: end function
14: function LOGSOFTMAX(logits)
15:   softmax_output  $\leftarrow \text{Softmax}(\text{logits})$ 
16:   return  $\log(\text{softmax\_output})$ 
17: end function
18: for  $i = 1$  to  $n$  do
19:    $O'_{T_i} \leftarrow \text{Softmax}(O_{T_i})$ 
20:    $O'_{S_i} \leftarrow \text{LogSoftmax}(O_{S_i})$ 
21:    $L_{\text{ST}} \leftarrow L_{\text{ST}} + (O'_{T_i} + O'_{S_i})$ 
22:    $L_{\text{MSE}} \leftarrow L_{\text{MSE}} + (O_{L_i} - O_{S_i})^2$ 
23: end for
24:  $L_{\text{ST}} \leftarrow \frac{L_{\text{ST}}}{n}$ 
25:  $L_{\text{MSE}} \leftarrow \frac{L_{\text{MSE}}}{n}$ 
26:  $L_{\text{Distillation}} \leftarrow \alpha \cdot L_{\text{ST}} + \beta \cdot L_{\text{MSE}}$ 
27: return  $L_{\text{Distillation}}$ 

```

IV. EXPERIMENTS

A. Datasets specifications

Human 3.6M: The Human3.6M dataset stands as the most extensive benchmark for 3D human pose estimation accessible to the public. Comprising 3.6 million frames, it was captured using four synchronized 50Hz digital cameras. Precise 3D coordinates of human joints were acquired through a marker-based motion capture system, further enhanced by 10 additional IR sensors. This dataset encompasses a diverse pool

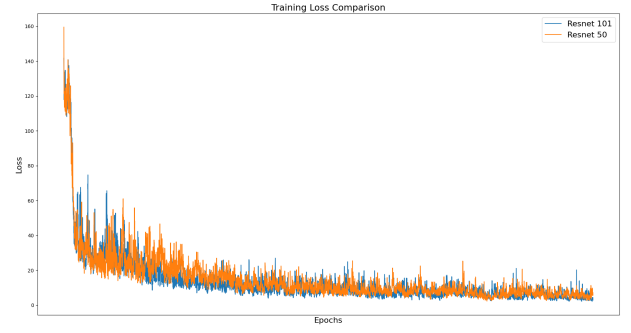


Fig. 2. Loss Trend Graph of each experimental group

of 11 subjects, including 5 females and 6 males, performing totally 15 different activities. For evaluating, we have followed the most popular protocol, by training on subjects images from folders numbers following as - 1, 5, 6, 7, 8 and using unseen subjects images from folders 9, 11 for testing. Similar to other methods, we use all available views during training and inference.

B. Training

Based on [14], we apply the main method of 3D pose estimation to train our Teacher Model to have a standard of basis to realize our knowledge distillation. Further more, cooperating with our approaches to conduct knowledge distillation which decreases the count of parameters and forms a new Model named Student Model, we finally got the version of Student Model after 7500 epoches. For more details, our Teacher Model apply the ResNet-152 with total number of parameters - **61,106,458** produced fully during training process. After knowledge distillation conducted to Teacher Model with testing ResNet-50, ResNet-101 separately, the count of parameters of Student Models for each different types of ResNet can be decreased to **25,325,786** for ResNet-50 and **44,387,546** for ResNet-101. So that the rate of decreasing parameters for each types of ResNet from Teacher Model to Student Model can reach to 41% for ResNet-50 and 73% for ResNet-101.

C. Result

This is the specification that we use in the experiment process.

- CPU: 16 vCPU Intel(R) Xeon(R) Platinum 8352V CPU @ 2.10GHz
- GPU: RTX 4090 * 1

Knowledge distillation training loss

	1000	2000	5000	7500
ResNet-50	11 ± 2	13 ± 4	7 ± 1	6 ± 1
ResNet-101	30 ± 5	10 ± 2	7 ± 2	3 ± 1

Accuracy Retention

Model	Parameters Reduced	FPS
Teacher (ResNet-152)	-	22 FPS
Student (ResNet-50)	41%	58 FPS
Student (ResNet-101)	73%	37 FPS

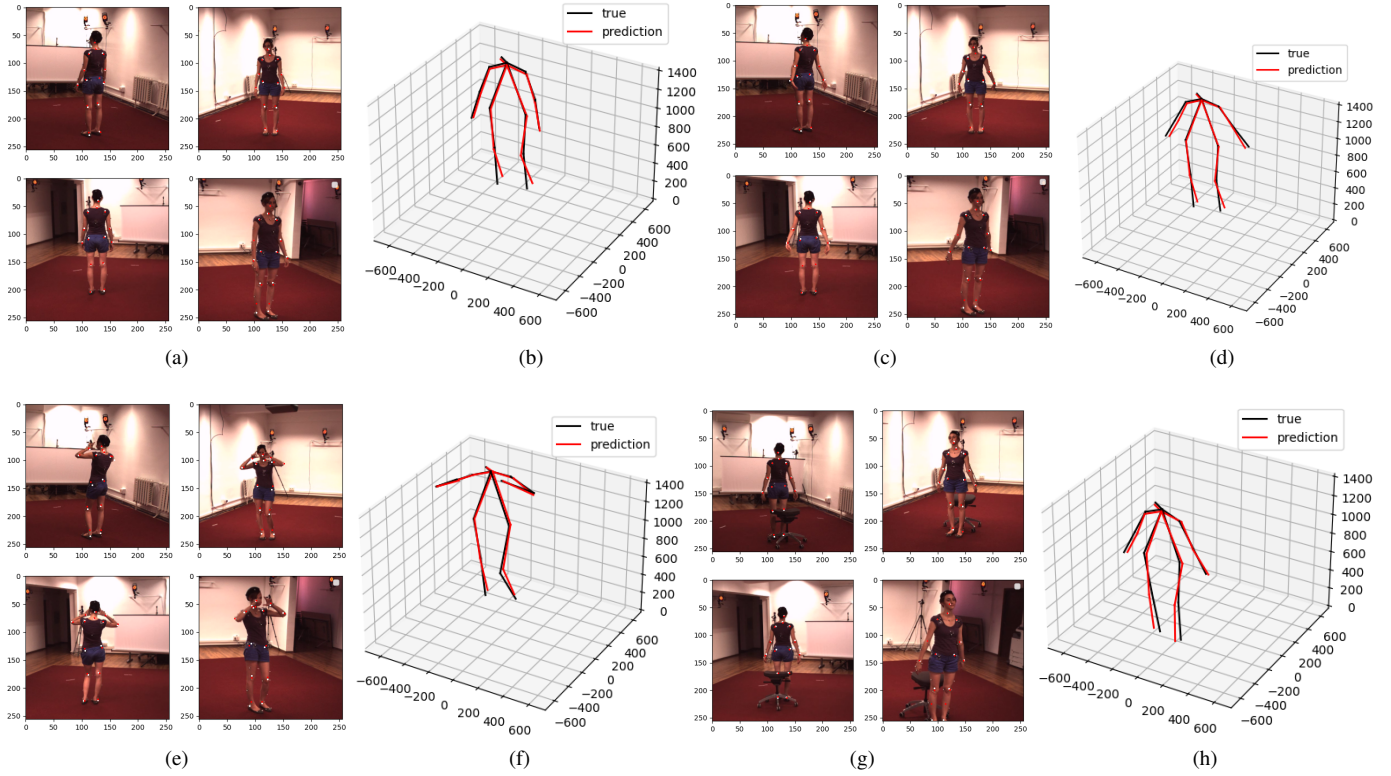


Fig. 3. Experiment Results on Human 3.6m

D. Analysis

The primary outcome of KD is the creation of small models. These models, distilled from their larger counterparts, possess much of the learned knowledge but in a more compact form. Small models are inherently advantageous because they require less memory and are easier to deploy in resource-constrained environments, such as mobile devices or embedded systems. One of the impressive benefits of using KD to train small models is the reduction in the number of training epochs required. Since the student model is guided by the pre-trained teacher model, it converges more quickly compared to training from scratch. The teacher model provides "soft targets" that contain rich information about the output distribution, which helps the student model learn more efficiently. Consequently, the training process is accelerated, requiring fewer epochs to achieve satisfactory performance. Reduced model size and fewer training epochs translate directly into lower computational power requirements. Training deep neural networks is typically computationally intensive, consuming significant time and resources. However, with KD, the smaller student models are less demanding in terms of computational power. This efficiency is particularly beneficial when scaling models across multiple systems or when deploying them in environments with limited computational capabilities.

By referring to the training result table as shown above which illustrate the Distillation Loss of models, for ResNet-50 Student Model, epochs of 1000 times seems to have less

loss which means higher accuracy than 2000 times of epoches. This situation can represent that for epoches ranging from 0-1000 and 0-2000, the 2000 times of epoches may reach the overfitting of model training. But the loss also continue to decrease over 5000 times of epoches. For ResNet-101 the loss is decreasing continuously when epoches reach from 1000 to 2000, 5000 and 7500 and can finally have 3 ± 1 loss meaning comparatively and efficiently accurate the Student Model with ResNet-101 is. What's more, for the first 50 epochs around, we can know that all kinds of the ResNet reduce the loss fast with significant efficiency. However, with the counts of epochs growing up like 5000 epoches or more, it seems that the ResNet-50 can be earlier to get stable and near to the lower loss point than ResNet-101 which means in the context of Knowledge Distillation (KD), models with fewer parameters, such as ResNet-50, tend to achieve better convergence and stability compared to models with a larger number of parameters, like ResNet-101.

V. CONCLUSION

In conclusion, our research has demonstrated the effectiveness of knowledge distillation in addressing the challenges associated with deploying 3D Human Pose Estimation (3D HPE) models in real-world scenarios. By leveraging knowledge distillation, we have successfully reduced the complexity and computational demands of the models while preserving their high accuracy.

Our exploration into the application of knowledge distillation for multi-view 3D human pose estimation has yielded promising results. We have shown that by distilling essential information from complex teacher models to lightweight student models, we can create streamlined and efficient 3D HPE models capable of meeting the stringent resource constraints of real-world deployment.

Moving forward, we envision our research serving as a foundation for further exploration and development in the field of 3D pose estimation and beyond. By empowering researchers and practitioners with adaptable and efficient tools, we aim to shape the landscape of computer vision applications, ultimately enhancing human-computer interaction, augmented reality, and other related fields.

VI. LIMITATIONS AND FUTURE WORK

While our research has made significant strides in addressing the challenges of deploying 3D Human Pose Estimation (3D HPE) models in resource-constrained environments through knowledge distillation, there are limitations that warrant further exploration. One limitation is the potential trade-off between model size and accuracy; although we have achieved a substantial reduction in parameters, there might be scenarios where even higher accuracy is required, possibly necessitating a reevaluation of the distillation process to preserve more information from the teacher model. Additionally, the generalization of our student models across various datasets and poses remains a challenge, as real-world applications may encounter poses and environments not represented in the Human3.6M dataset.

For future work, we envision several directions to build upon our current findings. Firstly, we aim to enhance the generalization capabilities of our models by incorporating a wider range of datasets, including those with varying poses, lighting conditions, and backgrounds. This will enable the student models to better adapt to diverse real-world situations. Secondly, we plan to explore adaptive knowledge distillation techniques that can dynamically adjust the distillation process based on the complexity of the input data, potentially improving both accuracy and efficiency. Furthermore, the integration of our models with other computer vision tasks, such as object recognition and scene understanding, presents an exciting avenue for research, as it could lead to more holistic and context-aware human-computer interactions.

Lastly, we are interested in the potential of transfer learning and few-shot learning to further reduce the dependency on large annotated datasets, making 3D HPE more accessible for applications with limited data availability. By pursuing these avenues, we hope to continue pushing the boundaries of what is possible in the field of computer vision and contribute to the advancement of human-centric AI technologies.

REFERENCES

- [1] A. Toshev and C. Szegedy, "DeepPose: Human Pose Estimation via Deep Neural Networks," in 2014 IEEE Conference on Computer Vision and Pattern Recognition, June 2014.
- [2] W. S. Low, K. Y. Goh, S. K. Goh, C. H. Yeow, K. W. Lai, S. L. Goh, J. H. Chuah, and C. K. Chan, "Lower extremity kinematics walking speed classification using long short-term memory neural frameworks," *Multimedia Tools and Applications*, vol. 82, no. 7, pp. 9745–9760, 2023.
- [3] S. K. Goh, N. P. Singh, Z. J. Lim, and S. Alam, "Interpretable tracking and detection of unstable approaches using tunnel gaussian process," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 3, pp. 2658–2671, 2023.
- [4] J. Carreira, P. Agrawal, K. Fragkiadaki, and J. Malik, "Human Pose Estimation With Iterative Error Feedback," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016.
- [5] B. Artacho and A. Savakis, "UniPose: Unified Human Pose Estimation in Single Images and Videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020.
- [6] Y. Chen, Z. Wang, Y. Peng, Z. Zhang, G. Yu and J. Sun, "Cascaded Pyramid Network for Multi-person Pose Estimation," 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, Salt Lake City, UT, USA, 2018, pp. 7103-7112, doi: 10.1109/CVPR.2018.00742.
- [7] J. Wang et al., "Deep High-Resolution Representation Learning for Visual Recognition," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 10, pp. 3349-3364, 1 Oct. 2021, doi: 10.1109/TPAMI.2020.2983686.
- [8] L. Pishchulin et al., "DeepCut: Joint Subset Partition and Labeling for Multi Person Pose Estimation," 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, 2016, pp. 4929-4937, doi: 10.1109/CVPR.2016.533.
- [9] K. He, G. Gkioxari, P. Dollár and R. Girshick, "Mask R-CNN," 2017 IEEE International Conference on Computer Vision (ICCV), Venice, Italy, 2017, pp. 2980-2988, doi: 10.1109/ICCV.2017.322.
- [10] L. Chen, H. Ai, R. Chen, Z. Zhuang and S. Liu, "Cross-View Tracking for Multi-Human 3D Pose Estimation at Over 100 FPS," 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2020, pp. 3276-3285, doi: 10.1109/CVPR42600.2020.00334.
- [11] J. Dong et al., "Fast and Robust Multi-Person 3D Pose Estimation and Tracking From Multiple Views," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 10, pp. 6981-6992, 1 Oct. 2022, doi: 10.1109/TPAMI.2021.3098052.
- [12] Z. Tang, Z. Qiu, Y. Hao, R. Hong and T. Yao, "3D Human Pose Estimation with Spatio-Temporal Criss-Cross Attention," 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Vancouver, BC, Canada, 2023, pp. 4790-4799, doi: 10.1109/CVPR52729.2023.00464.
- [13] J. Zhang, Z. Tu, J. Yang, Y. Chen and J. Yuan, "MixSTE: Seq2seq Mixed Spatio-Temporal Encoder for 3D Human Pose Estimation in Video," 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, LA, USA, 2022, pp. 13222-13232, doi: 10.1109/CVPR52688.2022.01288.
- [14] E. Remelli, S. Han, S. Honari, P. Fua, and R. Wang, "Lightweight Multi-View 3D Pose Estimation Through Camera-Disentangled Representation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020.
- [15] G. Hinton, O. Vinyals, and J. Dean, "Distilling the Knowledge in a Neural Network," arXiv:1503.02531 [cs, stat], Mar. 2015, Available: <https://arxiv.org/abs/1503.02531>
- [16] S. Han, J. Pool, J. Tran, and W. J. Dally, "Learning both Weights and Connections for Efficient Neural Networks," arXiv:1506.02626 [cs], Oct. 2015, Available: <https://arxiv.org/abs/1506.02626>
- [17] S. Han, H. Mao, and W. J. Dally, "Deep Compression: Compressing Deep Neural Networks with Pruning, Trained Quantization and Huffman Coding," arXiv:1510.00149 [cs], Feb. 2016, Available: <https://arxiv.org/abs/1510.00149>
- [18] J. Gou, B. Yu, S. Maybank, and D. Tao, "Knowledge Distillation: A Survey," *Int. J. Comput. Vis.*, 2021, doi: <https://doi.org/10.1007/s11263-021-01453-z>.
- [19] L. Wang and K. -J. Yoon, "Knowledge Distillation and Student-Teacher Learning for Visual Intelligence: A Review and New Outlooks," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 6, pp. 3048-3068, 1 June 2022, doi: 10.1109/TPAMI.2021.3055564.